

COASTAL EROSION

Smashing against the edges of continents, waves slowly wear away their fringes, cutting them back to form coasts but leaving the continental shelves below sea level. The rocks of exposed shores are eroded to create caves, cliffs, and rocky reefs. The debris is swept along the coast to more sheltered places where it is dumped as shingle banks, sandy beaches, and mudflats. So in some places the coast is cut back by erosion, and in others it is extended by banks and beaches.



▲ CLIFFS

Where the land surface is well above the sea level, its fringes get eroded to create steep cliffs of bare rock. These often rise above rock platforms that extend out to the sea beneath the waves, forming the inshore edge of the continental shelf.



▲ CAVES AND ARCHES

The higher a rock is above sea level, the less it is affected by waves, so if it is strong it may survive to form the roof of a cave or even a rock arch. Eventually, however, erosion cuts away the rock supporting it, and it comes crashing down into the sea.



▲ STACKS

The stumps of collapsed rock arches often form isolated stacks that are cut off from the land at high tide. Since they are usually made of particularly hard rock, they often survive for many years, while the rest of the coast is steadily cut back by erosion. They are frequently colonized by breeding seabirds because they are safe places to nest.

▲ WAVE ENERGY

Breaking waves drive water into cracks in the rock, forcing it apart. This loosens the rock and it falls away, often creating caves. The caves then collapse, leaving sheer cliffs that are attacked by more waves. This erosion process is most violent on shores that are exposed to strong winds and big waves. Soft rocks also erode faster, creating bays between headlands of harder rock.



◀ SHINGLE AND SAND

The rubble that falls in the sea is tossed around and smashed up to form boulders, shingle, and sand. Big, heavy boulders stay where they are, but lighter stones and sand are carried away from exposed shores by the currents. They are swept into bays and tossed up on shore by the waves, forming banks and beaches that help protect the coast from violent storms.